ILHAM KHADAFI

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PROFILE

7th semester Statistics undergraduate (**GPA 3.24**) at **Universitas Indonesia** with strong interests in **AI Engineering** and **Data Science**. Solid foundation in developing end-to-end **machine learning pipelines** and applying **deep learning** architectures across **Computer Vision**, **Natural Language Processing (NLP)**, and **time series forecasting**. Hands-on experience constructing advanced models from scratch, complemented by academic leadership as a Teaching Assistant guiding **100+ students** in **Numerical Methods** and **Data Science**. Highly analytical, detail-oriented, and driven to design scalable AI solutions for tech-driven industries.

EDUCATION

Universitas Indonesia — Depok, Indonesia
Bachelor of Statistics — *Aug 2023 – Aug 2027 (Expected)*

- **GPA/IPK**: 3.24 / 4.00
- **Relevant Coursework:** Unstructured Data Analysis, Statistical Computing, Data Mining & Business Intelligence, Machine Learning
- **Finalist** - ACTION 2025 Data Mining Competition
- **Finalist** - FORTEX 5.0 Data Science Competition

PROJECTS

Multi-Horizon GOOGL Stock Return Prediction — 2026
ML, Time Series, Financial Engineering

- Compared **LightGBM**, **TFT**, and **iTransformer** across 1d/5d/20d horizons on a **157-feature dataset** (technical, SEC EDGAR fundamentals, FRED macro)
- Implemented **iTransformer from scratch in PyTorch** (ICLR 2024), with 106K params, 10x leaner than TFT, highest Sharpe at 1d (1.09) and 5d (1.83)
- Designed stationarity-aware feature engineering, availability masks for sparse fundamentals, and non-overlapping rebalance windows

Topic Modeling of Gojek App Reviews Using LDA — 2026
NLP, Unsupervised Learning

- Processed **225,000** Indonesian-language reviews with a custom 888-term **PySastrawi** pipeline, modeled 30K-document sample using **LDA** to identify 7 user concern topics
- Replaced perplexity with **Coherence Score (C_v)** for K selection, peaked at K=7 (0.462), validated via K=6 vs. K=7 qualitative comparison

Named Entity Recognition for Culinary – IndoBERT — 2025
Deep Learning, NLP — Finalist, ACTION 2025 Data Mining Competition

- Fine-tuned **IndoBERT Transformer** on 2,700+ annotated samples for culinary entity extraction, achieved **97.48%** validation accuracy on token-level NER

Toxic Release Prediction & Spatial Analysis — 2025
ML, Spatial Analysis — Finalist, FORTEX 5.0 Data Science Competition

- Built **Random Forest** regression model (RMSE: 2.14) for toxic release risk prediction, applied **K-Means** clustering for spatial waste management segmentation

WORK EXPERIENCE

Teaching Assistant – Numerical Methods & Introduction to Data Science — Jul 2025 – Present
Department of Mathematics, Universitas Indonesia — *Depok, Indonesia*

- Mentored **100+ students** in computational methods and data science (Python, R), validated student code for practicum assignments and facilitated laboratory sessions

SKILLS

- **Programming:** Python (pandas, NumPy, statsmodels, matplotlib), R, SQL
- **AI & Machine Learning:** Supervised/Unsupervised Learning, Deep Learning, Computer Vision, NLP (Transformers, LDA), Time Series Forecasting
- **Models & Frameworks:** PyTorch, LangChain, OpenCV, XGBoost, LightGBM, CatBoost, Random Forest, Decision Trees, TFT, iTransformer, IndoBERT, scikit-learn
- **Tools:** Microsoft Office (Word, Excel, PowerPoint, etc.), VS Code, Git, Power BI
- **Languages:** Bahasa Indonesia (Native), English (Professional Working Proficiency)